

# Experiment - 03

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**(A)**

You are given with the Employee schema; the task is to find the highest score for each respective department.

For eg: department D1 have the highest score as 5.28. Hence in the output, final updated score for D1 should be 5.28.

Same with the other departments i.e., D2, D3 and D4.

OUTPUT:

| EID | DEPT | SCORES |
|-----|------|--------|
| 1   | D1   | 1      |
| 2   | D1   | 5.28   |
| 3   | D1   | 4      |
| 4   | D2   | 8      |
| 5   | D1   | 2.5    |
| 6   | D2   | 7      |
| 7   | D3   | 9      |
| 8   | D4   | 10.2   |

| EID | DEPT | SCORES |
|-----|------|--------|
| 1   | D1   | 5.28   |
| 2   | D1   | 5.28   |
| 3   | D1   | 5.28   |
| 4   | D2   | 8      |
| 5   | D1   | 5.28   |
| 6   | D2   | 8      |
| 7   | D3   | 9      |
| 8   | D4   | 10.2   |

**Solution:**

```
UPDATE EMPLOYEE AS E1
SET SCORES =
(
  SELECT MAX(SCORES)
  FROM EMPLOYEE
  WHERE DEPT = E1.DEPT
```

);

SELECT \* FROM EMPLOYEE;

**(B)**

A company maintains records of products sold by different salespersons. The management wants to prepare a sales performance report by identifying the salesperson(s) who generated the highest overall sales revenue. If multiple salespersons have the same highest total sales amount, all of them must be included in the final report. Return the seller ID(s) satisfying this condition.

Output

| seller_id |
|-----------|
| 3         |

**Solution:**

```
SELECT SELLER_ID
FROM ORDERS
GROUP BY SELLER_ID
HAVING SUM(AMOUNT) = (
    SELECT MAX(MAX_AMT)
    FROM
    (
        SELECT SELLER_ID, SUM(AMOUNT) AS MAX_AMT
        FROM ORDERS
        GROUP BY SELLER_ID
    )
)
```

**(C)**

A company maintains employee salary records for different departments. The management wants to prepare a report containing employees whose salaries fall among the top three highest distinct salary values within their respective departments. If multiple employees receive the same qualifying salary, all such employees must be included in the final report. Display the department name, employee name, and salary for every qualifying employee.

SalesPerson

| seller_id | name | city        | commission |
|-----------|------|-------------|------------|
| 1         | John | New York    | 15         |
| 2         | Amy  | Los Angeles | 13         |
| 3         | Mark | Chicago     | 12         |
| 4         | Pam  | Boston      | 15         |

Orders

| order_id | seller_id | amount |
|----------|-----------|--------|
| 1        | 1         | 1200   |
| 2        | 1         | 800    |
| 3        | 2         | 2500   |
| 4        | 3         | 1500   |
| 5        | 2         | 700    |
| 6        | 3         | 2000   |
| 7        | 4         | 3000   |
| 8        | 4         | 200    |

Company

| com_id | name   | city     |
|--------|--------|----------|
| 1      | RED    | Boston   |
| 2      | ORANGE | New York |
| 3      | YELLOW | Boston   |
| 4      | GREEN  | Austin   |

| ID | Name  | Salary | Departmentid |
|----|-------|--------|--------------|
| 1  | Joe   | 85000  | 1            |
| 2  | Henry | 80000  | 2            |
| 3  | Sam   | 60000  | 2            |
| 4  | Max   | 90000  | 1            |
| 5  | Janet | 69000  | 1            |
| 6  | Randy | 85000  | 1            |
| 7  | Will  | 70000  | 1            |

| ID | Name  |
|----|-------|
| 1  | IT    |
| 2  | Sales |

Output

| Department | Employee | Salary |
|------------|----------|--------|
| IT         | Max      | 90000  |
| IT         | Joe      | 85000  |
| IT         | Randy    | 85000  |
| IT         | Will     | 70000  |
| Sales      | Henry    | 80000  |
| Sales      | Sam      | 60000  |

## Solution:

```

SELECT D.NAME AS DEPARTMENT, E.NAME AS EMPLOYEE, E.SALARY
AS SALARY
FROM
EMPLOYEE AS E
JOIN
DEPARTMENT AS D
ON E.DEPARTMENTID = D.ID
WHERE
(
    SELECT COUNT(DISTINCT E2.SALARY)
    FROM EMPLOYEE AS E2
    WHERE E2.DEPARTMENTID = E.DEPARTMENTID
    AND

```

```
        E2.SALARY > E.SALARY  
) < 3  
ORDER BY D.NAME;
```